

Internal Research Report: Do Temporal Network Signals Predict How Scientific Concepts Spread Across Disciplines?

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1 Goal and Setup

This run investigated whether the early structure of a scientific concept’s topic cooccurrence network anticipates its later cross-disciplinary breadth, beyond simple measures of volume and reach. Two research questions guided the work:

RQ1 Which of 30–50 early temporal knowledge-network indicators (new edges, neighbourhood novelty, centrality change, community transitions, brokerage, clustering) predict size-adjusted cross-field breadth, validated on held-out field groups and onset cohorts?

RQ2 Through which routes do concepts spread between fields? Does the principle of relatedness apply: do concepts spread next to fields related to the ones currently retaining them?

The primary data source was the OpenAlex bulk snapshot (2026-09-23; 476,196,327 works; 129.4 million base works 1995–2022). Concepts were identified by Aho–Corasick title matching of 56,643 legacy OpenAlex concepts (levels 2–5) plus Wikidata aliases, with stemmed verification and an LLM precision gate (≥ 0.80). The final panel comprised 12,499 concepts and 27,393 concept-by-field adoption episodes across 26 venue fields.

Table 1: Panel composition. DEV concepts have home fields in Computer Science, Engineering, Biochemistry/Genetics/Medicine. Held-out groups cover the remaining field families. Source: Experiment 5, iteration 2.

Split	Concepts	Episodes
DEV (CS/Eng/BGM/Med, onset 2003–2009)	4,771	9,079
Cohort (onset 2010–2014, all fields)	4,356	9,799
Held-out Physical Sciences	742	1,662
Held-out Life & Environment	1,113	3,099
Held-out Social Sciences	1,352	3,320
Held-out Math & Decision Sci.	165	434
Total	12,499	27,393

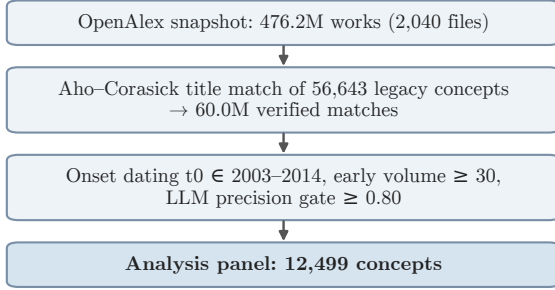
The primary outcome was *rarefied field breadth* O_{2r} ($m = 50$): the expected number of distinct venue fields among a fixed-size random draw of m papers from a concept’s publications in years $t_0 + 6$ to $t_0 + 8$, computed by exact hypergeometric rarefaction.

2 Iteration 1: Wide Screen of Five Rival Mechanisms

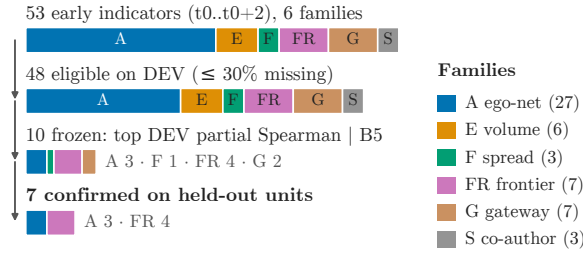
2.1 Why This Iteration Ran

The initial hypothesis centred on the background-adjusted naturalisation gap A_h^* , but a prior review computed its reliability at approximately 0.32, only 1 of 8 CIs excluded zero, and the aggregate off-home collapse did not match the stated mechanism. Rather than spending the whole iteration deepening a likely uninformative null, the strategy commissioned a wide screen of five rival mechanisms on one frozen dev panel (48 concepts, CS/Eng/BGM/Med homes, onset 2003–2009): lineage naturalisation (A_h^*), social reach through unconnected author groups, structural diversity of new cooccurrence neighbours, frequency-free selectivity, and landing in gateway fields. Five artifacts were commissioned; three completed and two failed.

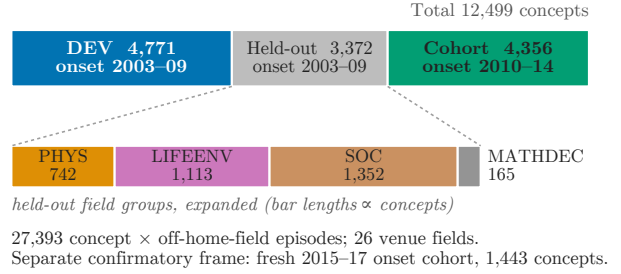
(a) Concept identification



(c) Indicator screen (breadth outcome)



(b) Panel structure



(d) Research questions

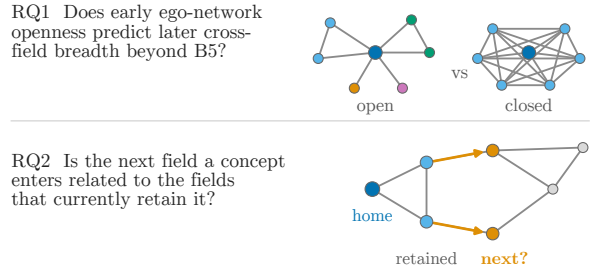


Figure 1: Overview of the study design. (a) Concept identification: 56,643 legacy OpenAlex concepts are matched by title (Aho–Corasick) against a 476.2M-work OpenAlex snapshot, giving 60.0M verified matches; concepts are onset-dated ($t_0 \in 2003-2014$), required to have early volume ≥ 30 and to pass an LLM precision gate ≥ 0.80 , yielding an analysis panel of 12,499 concepts. (b) Panel structure, bar lengths proportional to concept counts: DEV (blue; CS, Eng, BGM and Med homes, onset 2003–09) 4,771, held-out field groups (grey; onset 2003–09) 3,372, and the 2010–14 onset cohort (green) 4,356. The expanded lower bar splits the held-out block into PHYS 742, LIFEENV 1,113, SOC 1,352 and MATHDEC 165. The panel contains 27,393 concept $\times$ off-home-field episodes over 26 venue fields; a separate fresh 2015–17 onset cohort of 1,443 concepts serves as a confirmatory frame. (c) Indicator screen for the breadth outcome (O2r_m50). Bars are coloured by indicator family with lengths proportional to indicator counts: 53 early indicators computed over $t_0..t_0+2$; 48 eligible on DEV; the 10 frozen on DEV by partial Spearman correlation given the five-feature baseline B5; and the 7 confirmed on held-out units. (d) The two research questions as schematic illustrations.

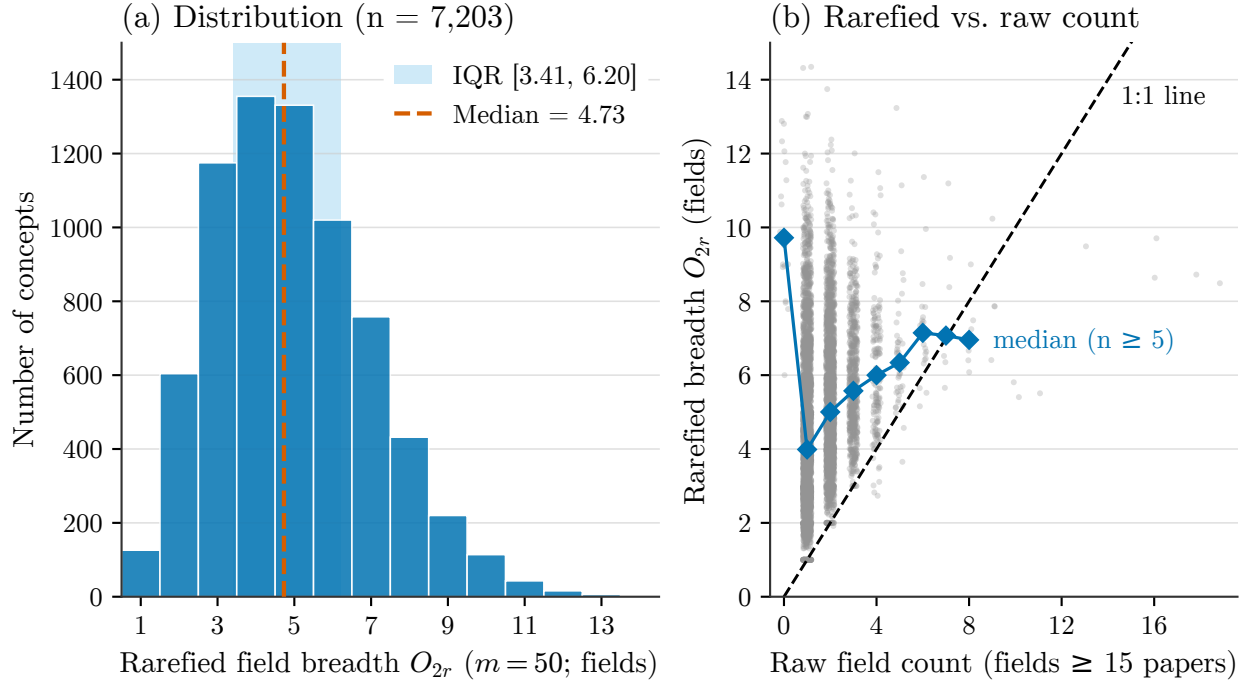


Figure 2: Distribution of the primary outcome, rarefied field breadth (O_{2r} , $m = 50$), over the 7,203 of 12,499 frame concepts that have at least 50 papers in the $t_0+6..t_0+8$ outcome window. (a) Histogram with the dashed orange line marking the median (4.73) and the light-blue band the interquartile range [3.41, 6.20]. (b) O_{2r} against the raw field count. Grey dots are concepts (x jittered), blue diamonds are per-count medians, and the dashed black line is the 1:1 line. Rarefied breadth is nearly independent of outcome-window volume ($\rho = -0.06$), whereas the thresholded raw count grows with volume ($\rho = 0.41$).

2.2 Experiment 1: Does citing a concept ‘as your own’ predict its spread?

Artifact workspace: `gen_art_experiment_1`.

Experiment 1 screened the background-adjusted naturalisation gap A_h^* on 48 dev concepts. The shared OpenAlex credit pool was exhausted (139 credits spent), so yearly counts came from OpenAlex exactly, while field labels, concept papers, citation lineage and background-reference fields came from free Semantic Scholar data.

Result: A_h^* does NOT survive the pre-registered rule. Leave-one-group-out $\Delta\rho$ for O_{2r} over $B5 = -0.006$, 90% concept-bootstrap CI $[-0.034, 0.017]$; $\rho_{B5} = 0.834$. Positive left-out groups: 0 of 4. Split-half reliability (Spearman–Brown) = 0.58 (below the 0.6 bar). None of the 14 candidate and foil features beat B5.

A methodological finding: R^2 of the raw concept-lineage log-odds ratio on the background log-odds ratio was 0.66. The background log-odds ratio was positive for 48/48 concepts — every field’s citing papers preferentially cite their own field above chance. Raw lineage is mostly disciplinary homophily.

Table 2: Pre-registered clause results for candidate A_h^* on 48 DEV concepts. Source: `gen_art_experiment_1`.

Clause	Value	Pass?
$\Delta\rho \geq 0.10$ & CI low > 0	$-0.006 [-0.034, 0.017]$	No
Positive groups $\geq 3/4$	0 of 4	No
Reliability ≥ 0.6	0.584	No
$ \rho_{\text{size}} \leq 0.6$	0.145 / -0.177	Yes

Dead end. A_h^* failed every clause. Closed as a headline bet.

2.3 Experiment 3: Do diverse topic ties predict concept spread?

Artifact workspace: `gen_art_experiment_3`.

Experiment 3 screened two cooccurrence emergence indicators on 47 dev concepts using the full-corpus topic PMI backbone (Leiden $\gamma = 3$, 25/26/23 communities per slice). The primary indicator D_ratio (the fallback for D_z, which failed the size diagnostic with $\rho = -0.63$ against log volume) gave $\Delta\rho = +0.006$, 90% CI $[-0.092, 0.135]$, 3/4 groups positive, Spearman–Brown 0.83. F_res gave $\Delta\rho = -0.060$, $[-0.158, 0.014]$, 1/4 groups positive, SB 0.44. Neither candidate survived the pre-registered rule.

An exploratory partial association of D_ratio given B5 was $+0.335 [0.02, 0.65]$, permutation $p = 0.037$, but with 12 tests this did not survive multiplicity correction. The portability table showed D_ratio, D_rare, participation and NOV_res associated with O_{2r} in all 4 groups (pooled ρ 0.45–0.63) but redundant under $\Delta\rho$ because B5 was already strong ($\rho_{B5} = 0.770$).

Dead end. No candidate survived. D_ratio carried forward only as the best available exploratory result.

2.4 Experiment 4: Where a concept lands early vs how broadly it spreads

Artifact workspace: `gen_art_experiment_4`.

Experiment 4 screened candidate G (gateway landing) on 46 dev concepts (34 with an outcome window). This artifact was also the authoritative producer of the shared outcome tables. B5 vs

B5+G on O_{2r} : $\Delta\rho = +0.033$, 90% CI $[-0.095, 0.168]$, positive in 2/4 groups, so G did NOT survive the pre-registered rule, although reliability ($r_{SB} = 0.92$) and the size check ($|\rho| \leq 0.13$) passed.

At the field level, the adopting field’s gateway (eigenvector) centrality added $\Delta\text{AUC} +0.103$ (95% CI $[0.03, 0.17]$) for retention, on 80 episodes. This became the iteration’s sole lead, although it was later disconfirmed at scale (Iteration 2).

Dead end (concept level). G failed the rule. The field-level lead was carried forward.

2.5 Failed Artifacts

Two artifacts failed with “REPL turn stalled”:

- **gen_art_dataset_1** (Sealed test set of new science concepts) — the outcome-blind held-out frame. Its failure meant no held-out evaluation existed after iteration 1.
- **gen_art_experiment_2** (Do independent author groups predict spread?) — candidate S, the Cheng et al. (2023) co-authorship hypothesis. Not run, not refuted at this stage.

2.6 Review and Hypothesis Update

The review scored iteration 1 at 3 (blocking). Key critiques: several stated conclusions contradicted the artifacts (the B5 ceiling argument did not apply to Exp4 where $\rho_{B5} = 0.327$; A_h^* medians were misread as correlation coefficients; “strongest secondary signal” was false). Two failed artifacts were missing from the record. The cross-experiment comparison was not like-for-like (each experiment computed its own O_{2r}).

The hypothesis update moved to **deepen** the sole lead: field-level gateway retention ($\Delta\text{AUC} +0.10$, dev only, $n = 80$). The headline moved from A_h^* (null) to the field-level gateway-retention lead. Evidence state: **lead**; confidence: **decreased**.

3 Iteration 2: Gateway Fields and Field Entry

3.1 Why This Iteration Ran

The field-level gateway-retention lead ($\Delta\text{AUC} +0.10$ on 80 dev episodes) was the only survivor from iteration 1. This iteration deepened it: building a 12,499-concept panel from the free OpenAlex S3 snapshot (zero API credits), adding confounds (field retention propensity, time-varying centrality, field FE), running a sealed held-out test, and testing the RQ2 field-entry mechanism. Five artifacts were commissioned; all five completed (Exp6 after a crash and re-run).

3.2 Experiment 5: Do hub fields keep new concepts? Held-out test

Artifact workspace: `gen_art_experiment_5`.

This artifact built the authoritative S1 panel (12,499 concepts, 27,393 episodes; Table 1) and tested H1: does the adopting field’s gateway centrality predict retention beyond B5, field size, relatedness to home, density, and the field’s leave-concept-out retention propensity?

Verdict: DISCONFIRMED. Held-out $\Delta\text{AUC} = -0.00001$ $[-0.0006, +0.0003]$ (DEV $+0.00001$). DerSimonian–Laird pooled -0.00004 ($I^2 = 0$). The minimum detectable ΔAUC was 0.004 at 90% power. Gateway centrality’s DEV signal ($+0.0019$ over the iteration-1 base) vanished once the field’s leaveconceptout retention propensity was added, and reversed on held-out (-0.0016). Gateway alone had AUC 0.605 on DEV vs. 0.506 on held-out: gateway was a domain-specific proxy for “fields that keep things.”

Within-field LPM: $\beta = +0.068/\text{SD}$, $p_{\text{concept}} = 0.041$ (two-way clustered $p = 0.17$). One pre-registered criterion passed, but the overall verdict remained DISCONFIRMED.

H3 (concept-level gateway landing predicting size-adjusted breadth): held-out partial ρ for G was 0.030, G_A 0.026, G_btw 0.046 (Holm $p = 0.0045$); DL pooled G 0.068 [0.029, 0.107]. The effect was small, and the concept-bootstrap CI of pooled G included zero ($[-0.006, 0.065]$).

Table 3: Gateway retention ladder: ΔAUC of gateway centrality over successive baselines for field retention on 27,393 episodes. Source: `gen_art_experiment_5`.

Rung	DEV ΔAUC	Held-out ΔAUC
L0 (field size only)	+0.0042	−0.0017
L1 (iter-1 base)	+0.0019	−0.0016
L2 (+ relatedness)	+0.0007	−0.0012
L3 (+ retention propensity)	+0.00003	−0.00004
L4 (full X_0)	+0.00001	−0.00001

3.3 Experiment 6: Where new scientific concepts spread next

Artifact workspace: `gen_art_experiment_6`.

On 653 newborn concepts (1,865 episodes), the conditional logit for next-field entry showed that relatedness to the off-home fields that currently *retain* the concept predicted the next field entered: held-out $d_0 = 0.302$ [0.240, 0.369], LR 71.7, DL pooled 0.284 [0.216, 0.352] ($I^2 = 0$), label-permutation $p = 0.001$, rewired-backbone $p = 0.015$. Verdict: **CONFIRMED** by the frozen rule.

However, the gateway weighting added nothing beyond plain retaining relatedness (M3 vs M1 $p = 0.17$ held-out). The trajectory typology (DTW $k = 2$) was stable (bootstrap ARI 1.0) but dominated by Medicine homes in the localised class.

Dead ends: Rescue (re-import), relay and gateway weighting were NOT supported on held-out.

3.4 Evaluation 1: Stress-testing the gateway-field retention lead

Artifact workspace: `gen_art_evaluation_1`.

Zero-API stress test of the iteration-1 +0.103 ΔAUC . On the union panel (362 de-duplicated episodes, 54 concepts), $\Delta\text{AUC} = +0.001$ [−0.012, 0.012]. A shuffled- R placebo’s 95th percentile (0.130) exceeded the original +0.103, so the original lead could not be certified as above chance on 80 episodes. The O1 (uptake) gains of G variants were label-coverage artefacts (G +0.072 $\rightarrow$ +0.002 once `label_coverage_early` was controlled).

3.5 Dataset 2: When research concepts were officially recognised

Artifact workspace: `gen_art_dataset_2`.

External-recognition lookup table (outcome O5) for 65,026 OpenAlex legacy concepts, built with zero credits and \$1.49 of OpenRouter calls. Sources: MeSH 2026 (20,872 concepts), English Wikipedia creation dates (6,540 exact first revisions), Wikidata P571/P575 (1,425), ACM CCS, MSC, PACS/PhySH, curated lists (589). 67% of concepts were recognised at or before onset. O5 was later found unrelated to publication outcomes (ρ with $O_{2r} = 0.014$ [−0.045, 0.073]).

Table 4: Pre-registered verdict ladder for the gateway-retention lead. Source: `gen_art_evaluation_1`.

Verdict	Rule
FAILS	new-episodes-only panel ΔAUC over M2 ≤ 0
FIELD-TRAIT	P_{cj} alone carries the gain AND gateway adds ≤ 0.01 given P_{cj} ; OR gateway sits inside C2 node-label permutation null union AND new-episodes $\Delta\text{AUC} > 0$ with 95% CI > 0 , same sign $\geq 3/4$ groups, delta over M2+ P_{cj} CI > 0 , real delta above C2 95th percentile
REPLICATES	
ATTENUATES	positive point estimates on both panels but ≥ 1 REPLICATES condition fails

3.6 Research 1: How our results compare with related papers

Artifact workspace: `gen_art_research_1`.

Positioning study for the Applied Network Science paper. It identified 22 citable ANS papers, comparison numbers from Guevara 2016 (field-entry AUC 0.68–0.90 for individuals, different unit), and the ANS template (unstructured abstract, author-year citations). The “Networks for everyday life” collection page could not be read (Springer IdP/JS blocked).

3.7 Review and Hypothesis Update

Review score: 3 (blocking). The gateway-retention lead was honestly disconfirmed. The retained-field relatedness lead ($d_0 = 0.28$) from Exp6 became the new headline. The reviewer noted that the “common panel” design was not realised (Exp5 and Exp6 used different frames), and many iteration-1 corrections remained unapplied.

Hypothesis update: **deepen** the retained-frontier lead. The headline moved from gateway centrality (closed) to the retained-frontier lead. Evidence state: **lead**; confidence: **decreased**.

4 Iteration 3: The Indicator Screen and the Frontier Test

4.1 Why This Iteration Ran

Two goals: (1) the decisive frontier test — retained-field relatedness must beat the conventional RCA-thresholded Hidalgo/Guevara density on an independent frame; (2) the RQ1 held-out indicator screen on the 12,499-concept panel, addressing the largest scope gap. Five artifacts were commissioned; four completed and one (Experiment 9, RQ2 trajectories) failed.

4.2 Experiment 7: Do concepts spread from fields that keep them?

Artifact workspace: `gen_art_experiment_7`.

The decisive test of the retained-frontier claim, on an independent frame (Exp5 12,499 minus every Exp6 concept = 11,841). The spec was hash-frozen on DEV (4,486 concepts); held-out was scored once.

Held-out pooled (PHYS+LIFEENV+SOC+MATHDEC, 3,162 concepts, 6,978 entries): $d_0 = 0.322$ [0.291, 0.355], LR(R3 vs R2) = 325.8, concept refit bootstrap 1,000. Positive in PHYS 0.15, LIFEENV 0.40, SOC 0.30, MATHDEC 0.07 (underpowered). DL 4 groups: 0.243 [0.118, 0.368], $I^2 = 0.92$. Retained-label permutation $p = 0.001$, rewired $p = 0.004$, node-label $p = 0.003$.

Frozen verdict: FRONTIER = PARTIAL. The pre-declared volume-matched contrast (retained vs entered-not-retained fields in the same volume cell) was null: $-0.028 [-0.105, 0.046]$, so persistence could not be separated from sustained volume. Under Hidalgo’s minimum-conditional-probability proximity, d_0 vanished (-0.021 , $p = 0.012$) — backbone-specific. Dose-response by retention age: held-out $0.098/0.075/0.304$ (not monotone).

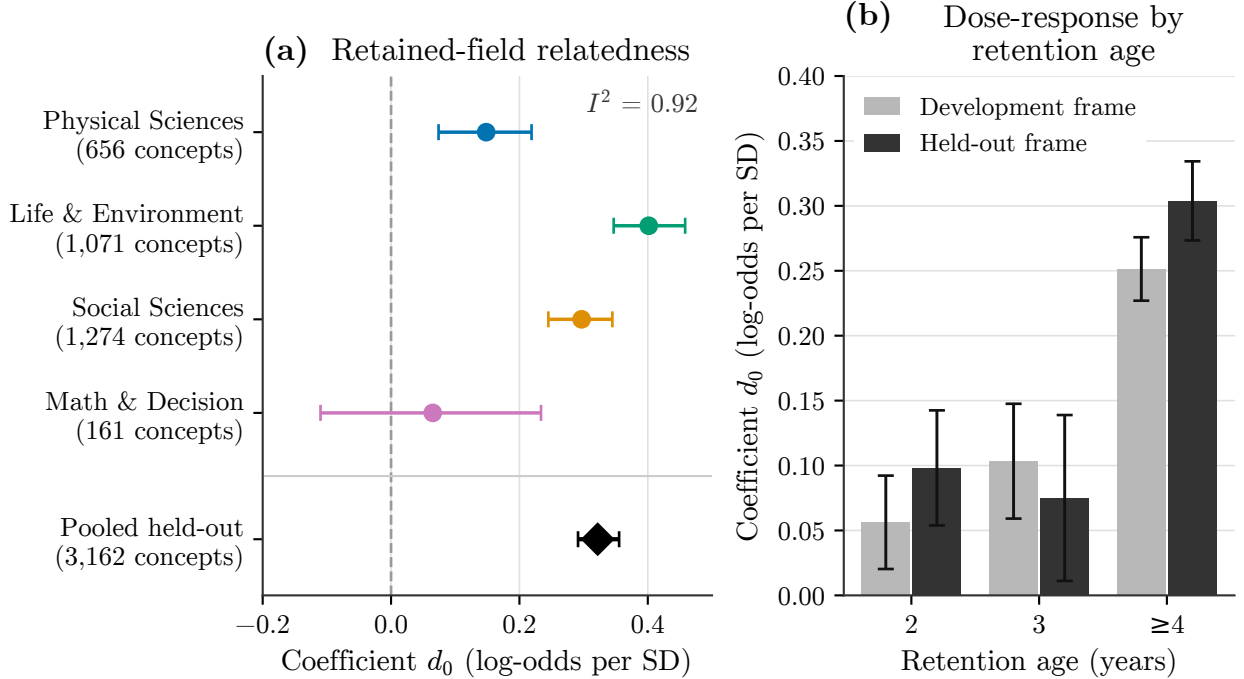


Figure 3: Retained-field relatedness predicts the next field a concept enters. (a) Conditional-logit coefficient d_0 on the held-out groups of the independent frame. Coloured circles are per-group estimates with 95% concept-bootstrap CIs. The black diamond is the pooled held-out estimate, $d_0 = 0.322 [0.291, 0.355]$ (3,162 concepts, 6,978 entry events). Three groups are clearly positive. The Math & Decision interval crosses zero, and the groups differ substantially ($I^2 = 0.92$). (b) d_0 split by retention age (2, 3, ≥ 4 years), for the development frame and the held-out frame, with 95% Wald CIs.

4.3 Experiment 8: Which early network signals of new topics travel

Artifact workspace: `gen_art_experiment_8`.

The RQ1 held-out deliverable on the 12,499-concept frame. 53 indicators in 6 families over $t_0..t_0+2$ (popularity E, disciplinary F, landing G, retained-frontier FR, 27 cooccurrence ego-network A, coauthor S) plus B5 baseline. Outcomes: O1c/O1b uptake, O2r breadth, O3 transience, O4 citation growth, O5 external recognition. DEV-only ranking, hash seal, single unseal, DL pooling, Holm correction.

Seven of 10 were confirmed with 6/6 unit sign agreement. The confirmed indicators spanned two families: retained frontier (M0_density_end, D_vol_end, CONTACT_REACH, RETENTION_RATIO) and cooccurrence topology (n_{comm} W3, NOV_{res}, ego_density W3). An ElasticNet combining all indicators added Spearman $+0.059 [+0.046, +0.073]$ over B5 on held-out groups; an EBM added $+0.052 [+0.037, +0.067]$.

Table 5: Held-out indicator screen: the 10 indicators with highest DEV partial Spearman priority for rarefied breadth (O_{2r} , $m = 50$), tested on held-out groups. DerSimonian–Laird pooled estimates with 95% CI. Source: Experiment 8, iteration 3.

Indicator	Family	Pooled PSP	95% CI	I^2	Sign	Confirmed
M0_density_end	FR	+0.375	[+0.279, +0.462]	0.74	6/6	Yes
D_vol_end	FR	+0.307	[+0.256, +0.356]	0.10	6/6	Yes
CONTACT_REACH	FR	+0.211	[+0.161, +0.261]	0.00	6/6	Yes
n_{comm} (W3)	A	+0.167	[+0.063, +0.267]	0.78	6/6	Yes
NOV _{res}	A	+0.151	[+0.044, +0.255]	0.75	6/6	Yes
RETENTION_RATIO	FR	−0.114	[−0.160, −0.067]	0.00	6/6	Yes
ego_density (W3)	A	−0.102	[−0.151, −0.053]	0.00	6/6	Yes
Rao–Stirling	A	−0.072	[−0.153, +0.010]	0.44	5/6	No
G_{btw}	G	+0.056	[−0.006, +0.118]	0.33	6/6	No
log offhome vol.	F	−0.089	[−0.171, −0.007]	0.63	5/6	No

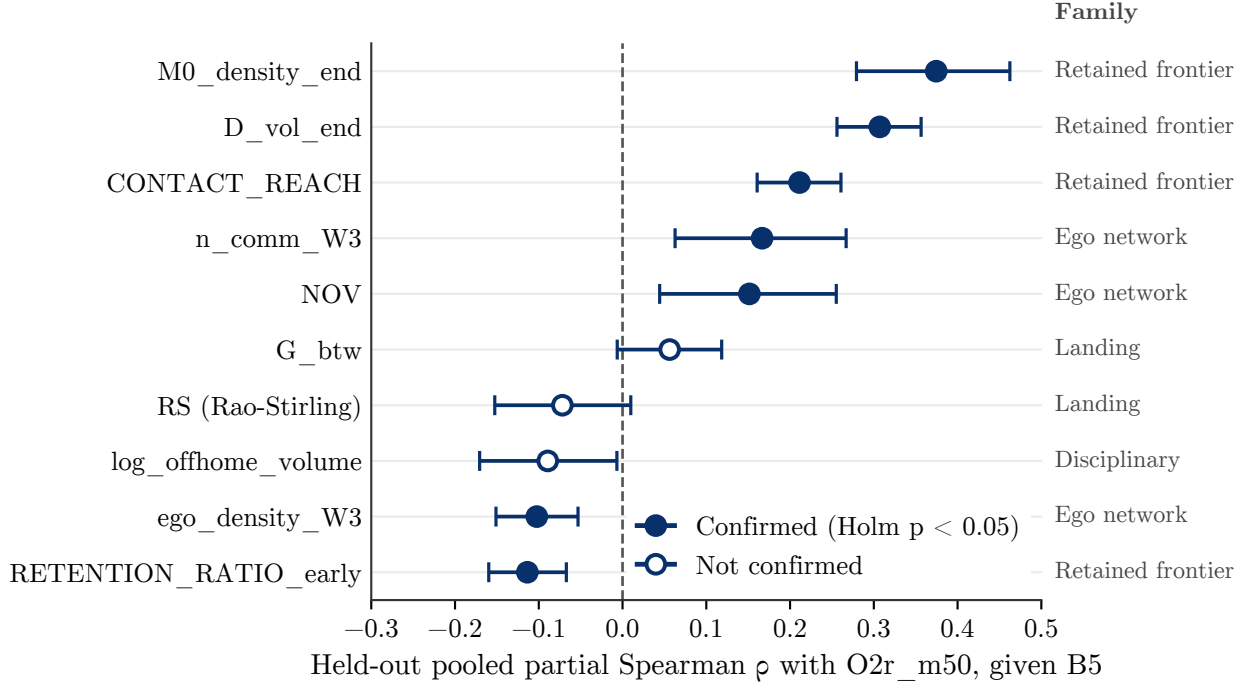


Figure 4: Held-out screen of the 10 indicators frozen on the DEV home groups for rarefied cross-field breadth (O2r_m50). Each row shows the DerSimonian–Laird pooled partial Spearman ρ with O2r_m50 given the B5 baseline. Filled circles mark confirmed indicators (Holm $p < 0.05$, pooled sign equal to the frozen DEV sign). Open circles mark indicators not confirmed. Seven of the ten are confirmed.

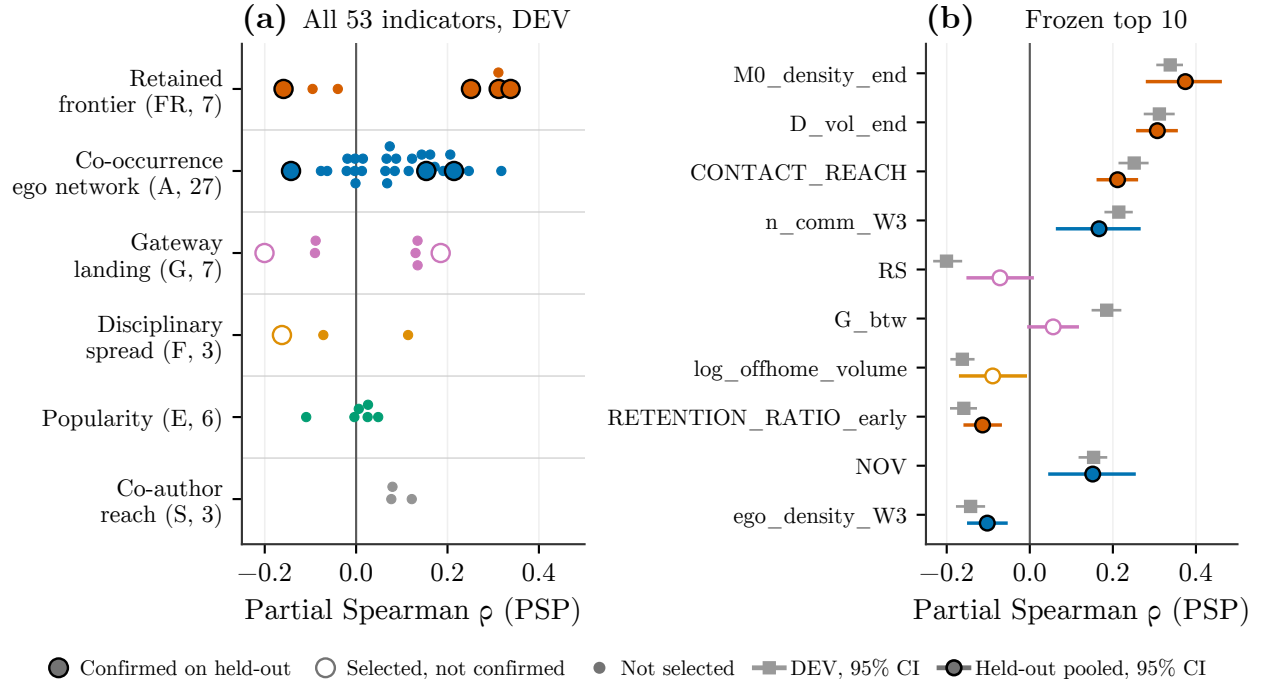


Figure 5: Complete indicator screen for rarefied cross-field breadth. (a) Partial Spearman ρ of all 53 screened indicators on the DEV fields, one row per family. Large filled circles are the 7 confirmed indicators; large open circles are the 3 selected on DEV but not confirmed; small dots are the 43 not selected. (b) The frozen top 10 in DEV rank order. Grey squares are DEV estimates; circles are held-out estimates. About half of the M0_density_end and D_vol_end effects reflects the concept's pre-onset field footprint.

Caveat: M0_density_end and D_vol_end use cumulative field history from 1995 to $t_0 + 2$, so part of their signal is a pre-onset footprint. O5 external recognition: no indicator or model beat B5 + onset year.

Pre-registered predictions: P2 (edge persistence negative) held; P1, P3, P4, P5 failed. Notably, P3 failed because new_edge_rate transferred to held-out (+0.118 [0.072, 0.163], 0 sign flips), reversing the iteration-1 dead end that “raw cooccurrence growth indicators fail to generalise.”

4.4 Evaluation 2: Auditing the record before the paper

Artifact workspace: `gen_art_evaluation.2`.

A zero-new-data audit of the record. 246 claims checked (224 MATCH, 15 MISLABELLED, 6 MISMATCH, 1 FILE_FLAG_OVERRIDDEN). Key findings: Dataset 2 source counts were external-entry counts not concept counts; 7 of 12 partial associations were available contrary to the report’s claim; O5 was unrelated to publication outcomes. The evaluation produced ready-to-paste corrections (`text_corrections.md`) and missing tables.

Table 6: Eval2 refit check: reproduced point estimates vs reported values across iterations 1–2. All reproductions matched exactly. Source: `gen_art_evaluation.2`.

Experiment	Outcome	Reported	Reproduced
Exp1	O_{2r}	−0.0056	−0.0056
Exp3 (D_ratio)	O_{2r}	+0.0060	+0.0060
Exp3 (F_res)	O_{2r}	−0.0604	−0.0604
Exp4 (G)	O_{2r} ($m=30$)	+0.0333	+0.0333
Exp4 (G)	O_{2r} resid	+0.1503	+0.1503
Exp4 (G)	O1	+0.0723	+0.0723
Exp4 (G_btw)	O1	+0.0163	+0.0163

4.5 Research 2: Prior art and venue check

Artifact workspace: `gen_art_research.2`.

Claim A (next-field entry follows relatedness to retaining fields): **PARTIALLY ANTICIPATED**. Persistence is used only as a filter on the entry outcome in prior work. No retained-only or duration-weighted density predictor was found. Claim B (relatedness to dropped fields lowers entry): mechanism partly anticipated; new as a test.

4.6 Failed artifact: Experiment 9

`gen_art_experiment.9` (“How new concepts spread: paths and reasons”) was commissioned but failed (output_format validation failed after 5 retries; `method.py` never ran). This was iteration 3’s entire RQ2 artifact: the log-additive decomposition, DTW+HMM typology on 12,499 concepts, sequence tests, case studies and O5 timing.

4.7 Review and Hypothesis Update

Review score: 3 (blocking). The record still contained mislabelled outcomes (O4 citation growth reported as transience) and unapplied Eval2 corrections. The Exp8 held-out screen was recognised as closing the largest scope gap.

Hypothesis update: the headline moved from the retained frontier (closed: volume-matched null, backbone-specific) to **openness vs consolidation**, built from the Exp8 held-out lead. Evidence state: **lead**; confidence: **decreased**.

5 Iteration 4: Fresh-Cohort Test and RQ2 Decomposition

5.1 Why This Iteration Ran

The Exp8 lead (six openness components predicted held-out breadth) needed replication on a never-screened cohort, a home-only build to control mechanical coupling, and confound controls (concept type, pre-onset footprint). RQ2 trajectories needed to be completed after Exp9’s failure. Five artifacts were commissioned; four completed. Experiment 11 (within-concept closure test) ran its DEV models but was interrupted before held-out scoring.

5.2 Experiment 10: Do open-neighbourhood concepts spread? Fresh-cohort test

Artifact workspace: `gen_art_experiment_10`.

Single-unseal confirmation of the RQ1 openness claim on a fresh 2015–2017 onset cohort (1,443 concepts after LLM precision gate; 573 with $\text{OPEN}_{\text{home}}$). Pre-seal power was 0.16. The OPEN index was the mean of six signed, z -scored ego-network components, with constants frozen from the 12,499 Exp5 concepts.

Table 7: OPEN index control ladder on the 2015–2017 confirmatory cohort. PSP = partial Spearman priority with rarefied breadth (O_{2r} , $m = 50$), 95% concept-bootstrap CI. Source: Experiment 10, iteration 4.

Build	R0	R1	R2	R3	R4	R5	n
$\text{OPEN}_{\text{home}}$	+0.12	+0.10	+0.09	+0.08	+0.07	+0.06	573
OPEN_{all}	+0.21	+0.18	+0.17	+0.17	+0.15	+0.14	630
$\text{OPEN}_{\text{size}}$	+0.18	+0.15	+0.15	+0.14	+0.12	+0.11	591

Verdict: CONFIRMED but marginal. $\text{OPEN}_{\text{home}}$ PSP with O_{2r} : +0.091 [+0.013, +0.171] at R2, +0.080 [+0.001, +0.162] at R3. CIs included zero at R4/R5. DL pool over groups: +0.083 [−0.007, +0.173], Holm $p = 0.048$. It added no practical prediction (B5 Spearman 0.768 vs. 0.770, +0.002 [−0.003, +0.008]). Mechanical coupling was large: OPEN_{all} +0.174, ALL minus HOME +0.093 [+0.016, +0.169].

5.3 Experiment 11: Does closing up at home slow a concept’s spread? (incomplete)

Artifact workspace: `gen_art_experiment_11` (failed/incomplete: no `.aii_worker_result.json`).

This artifact sealed a within-concept pre-registration and ran the DEV body models on 35,328 concept-years from 4,661 concepts. H-M1 density PPML $b = -0.070$ [−0.180, 0.040]; H-M2 $\text{OPEN}_{\text{home}}$ $b = +0.015$ [−0.038, 0.069]. By the frozen rule: **NOT SUPPORTED**. The Sun–Abraham event study was interrupted. Held-out, cohort and event study were not run. Later completed in iteration 5 (Experiment 15 part C).

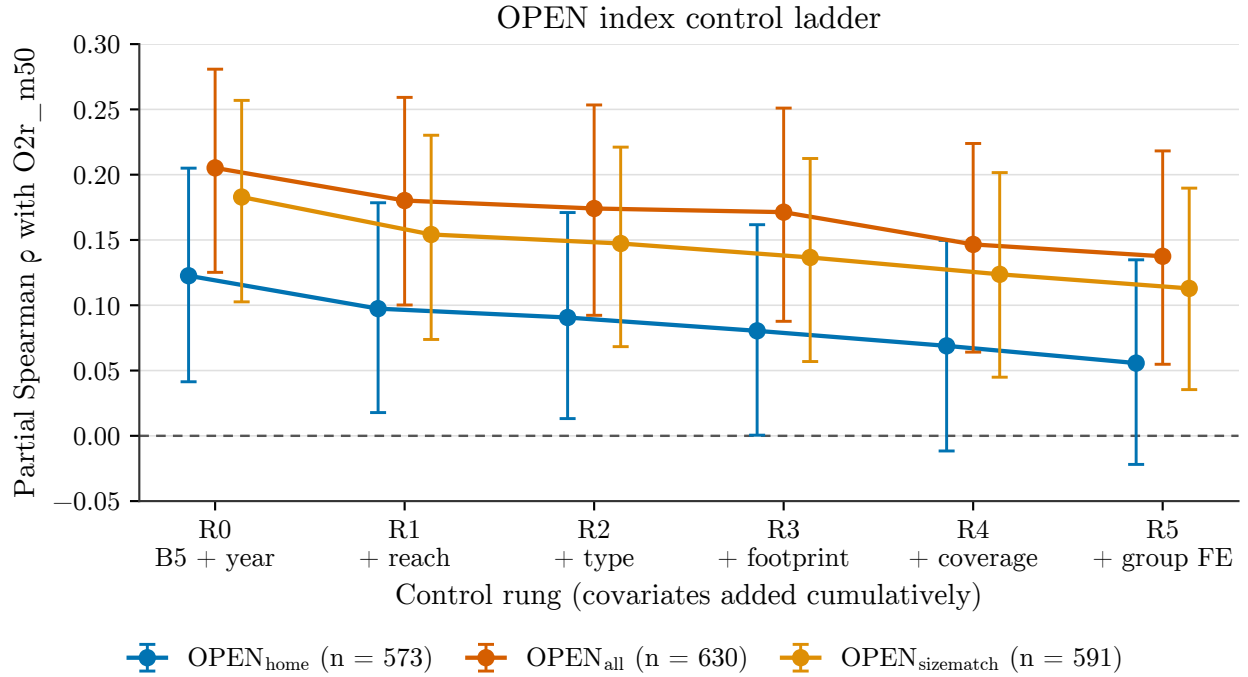


Figure 6: Control ladder for the OPEN index on the fresh 2015–2017 onset cohort. The x-axis gives cumulative control rungs: R0 = B5 + onset year, R1 = + contact reach, R2 = + concept type, R3 = + pre-onset footprint, R4 = + label coverage, R5 = + group fixed effects. Circles are point estimates; error bars are 95% concept-bootstrap CIs. OPEN_{home} (blue) is marginal: its CI touches zero at R3 and includes zero at R4 and R5.

5.4 Experiment 12: How concepts spread — contact versus keeping

Artifact workspace: `gen_art_experiment_12`.

The RQ2 trajectories analysis on all 12,499 Exp5 frame concepts. The breadth decomposition showed that broad concepts start with wider contact, not by advancing a wider frontier:

Table 8: Breadth decomposition ($\log B_n = \log E_2 + \log M + \log \rho$): share of the top-versus-bottom tercile gap in rarefied breadth attributable to each component. DEV concepts, variant i_pooled, $n = 3,188$. Source: Experiment 12, iteration 4.

Component	Share	95% CI
Exploration ($s_{E_2} + s_M$)	0.732	[0.703, 0.764]
Early contact diversity (s_{E_2})	0.779	[0.738, 0.818]
Retention (s_ρ)	0.268	[0.236, 0.297]
Difference (explore – retain)	0.464	[0.407, 0.528]

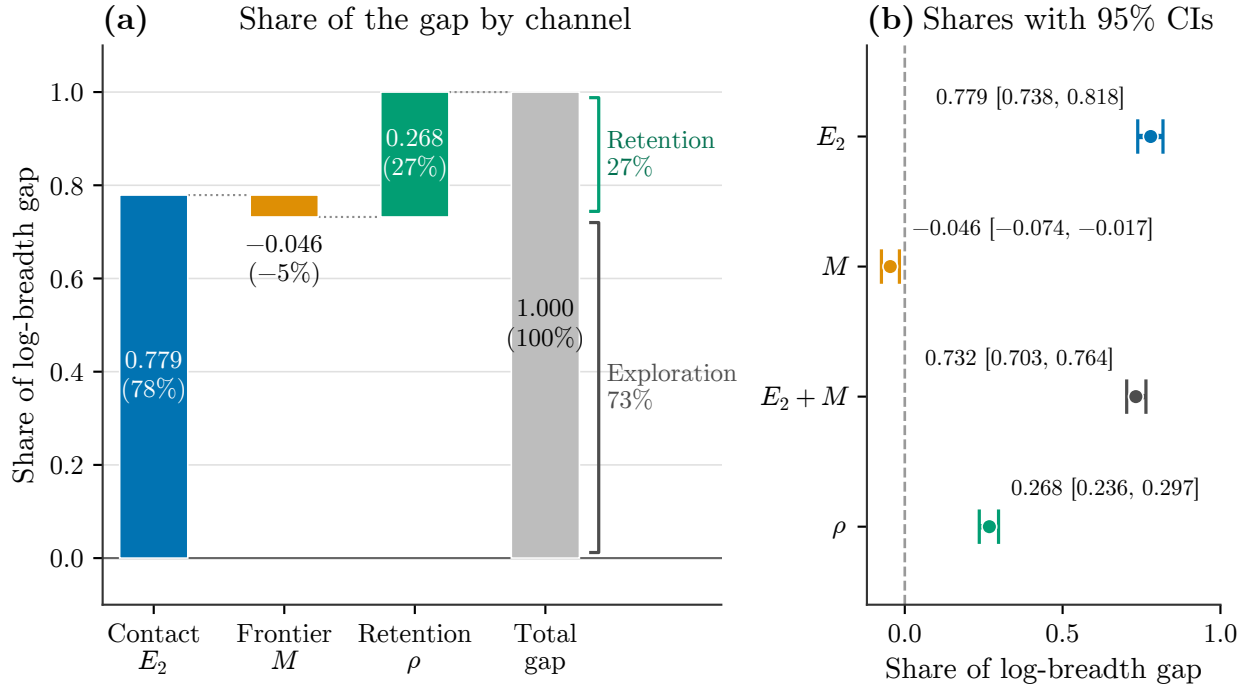


Figure 7: Log-additive decomposition of the gap in retained off-home breadth between the top and bottom terciles of size-adjusted breadth. DEV concepts, $n = 3,188$. Early contact diversity E_2 (blue) contributes 0.779 (78%), frontier advance M (amber) is slightly negative at -0.046 (-5%), and retention ρ (green) contributes 0.268 (27%). Broad concepts start with wider contact; they do not mainly retain more fields later.

The pre-registered variant iv (Medicine excluded) gave DEV 0.633 [0.537, 0.727], held-out pooled 0.492 [0.403, 0.575], cohort 0.445 [0.358, 0.527], DL 0.504 [0.329, 0.679], $I^2 = 0.76$. These shares are an accounting identity for the breadth outcome, not causal effects.

No trajectory typology passed the naming rule (DTW $k = 4$ vs HMM $S = 5$ ARI = 0.222; held-out re-cluster ARI 0.44/0.38). The result was **CONTINUUM**: PC1 38.8% (breadth-of-spread axis), PC2 10.7% (keep-vs-lose axis). OPEN tracked PC1, not PC2.

5.5 Evaluation 3: Record fixes and openness robustness tests

Artifact workspace: `gen_art_evaluation.3`.

This evaluation produced a corrections pack (76 APPLIED, 5 ALREADY_PRESENT) and a boundary study. Key finding: about half of the two biggest breadth effects was pre-onset footprint — `M0_density_end` dropped from +0.374 to +0.187 post-onset (attenuation 0.50), and `D_vol_end` from +0.317 to +0.176 (attenuation 0.45). A specification curve across 1,920 specifications showed 99.7% of pooled CIs above zero (median 0.152, Freedman–Lane $p = 0.005$), though this was labelled exploratory (all-papers build, previously unsealed groups).

5.6 Research 3: Is ‘keep exploring, spread widest’ new?

Artifact workspace: `gen_art_research.3`.

Novelty verdicts: C1 (openness → later breadth) **PARTIALLY ANTICIPATED** by Maillart 2026, Wang 2017, Weng 2013, Ugander 2012. C2 (consolidation → less breadth) **PARTIALLY ANTICIPATED** in mechanism; contradicted-by on other outcomes (Cheng 2023 consistency predicts volume). C3 (low retention ratio) **NEW**. C4 (within-concept closure → entry slowdown) **NEW** as a lead-lag test.

5.7 Review and Hypothesis Update

Review score: 2 (blocking). The case-study table contained fabricated rows. Experiment 11 was missing. Several conclusions contradicted their own artifacts (the coupling interpretation, OPEN vs the retention axis, PR2). Many Eval3 corrections remained unapplied.

Hypothesis update: claim sharpened from six-component “openness” to the decoupled home-only signal the cohort isolated: novel partners ($\text{NOV}_{\text{res}} +0.134$) and churn (edge_persistence -0.112). Evidence state: **lead**; confidence: **decreased**.

6 Iteration 5: Vocabulary-Free Confirmation and the Cheng Reversal

6.1 Why This Iteration Ran

The fresh-cohort lead was fragile (R4/R5 CIs included zero, DL CI included zero, pre-seal power 0.16, about half of the Exp8 signal was coupling). This iteration sought confirmation on a second, vocabulary-free population (Frame N) and tested the Cheng reach-vs-depth reversal, the sampling-noise confound, the mechanism (where new partners come from), and the Exp11 closure test completion. Five artifacts were commissioned; all five completed.

6.2 Experiment 13: Does the churn signal hold for brand-new phrases?

Artifact workspace: `gen_art_experiment.13`.

Frame N: 636 vocabulary-free phrase-born concepts (onsets 2003–2015) outside all legacy labels. Zero credits: two passes over the S3 snapshot.

Frozen verdict: **PARTIAL**. $\text{OPEN}_{\text{home}}$ PSP: +0.117 [+0.020, +0.218] at R3, +0.086 [−0.009, +0.190] at R5. 3/4 estimable groups positive (SOC −0.025); DL +0.112 [−0.015, +0.239]; Holm $p = 0.052$. NOV_{res} carried the signal (+0.208 [+0.113, +0.303]); edge persistence was null (−0.013). No forecasting gain over B5 (Spearman 0.80).

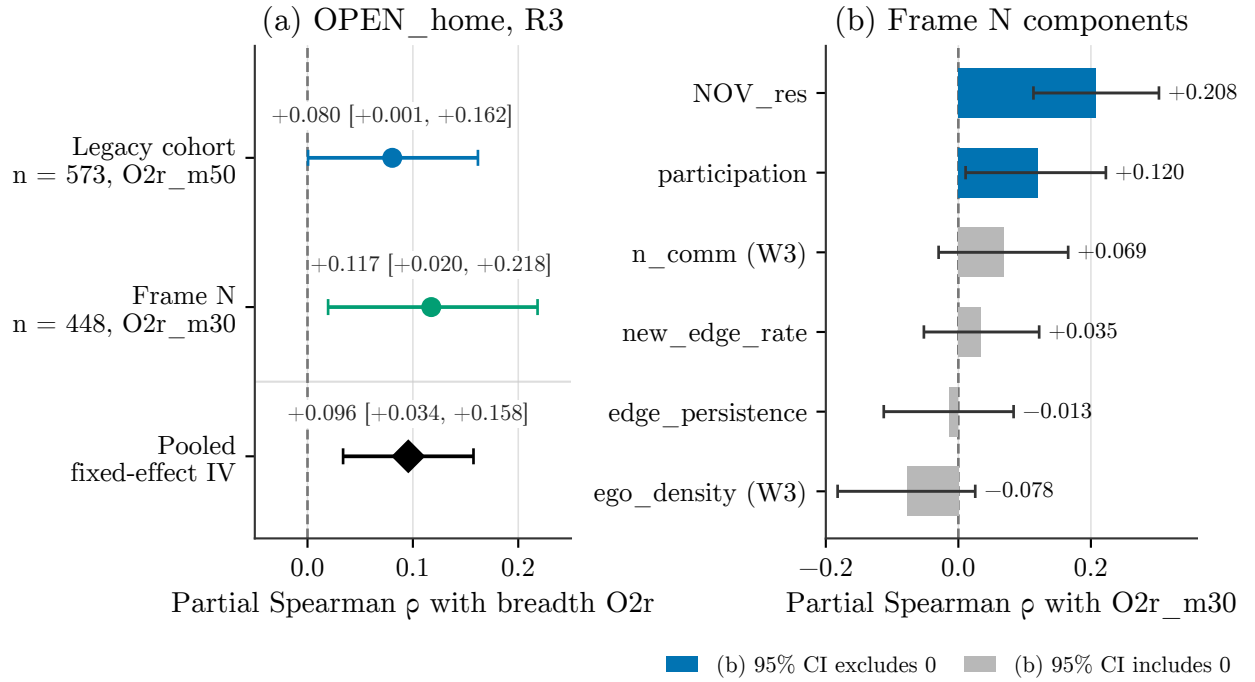


Figure 8: Confirmation of the OPEN signal on vocabulary-free Frame N concepts. (a) Forest plot of OPEN_{home} PSP with rarefied cross-field breadth at the footprint rung (R3): the 2015–17 legacy cohort ($n = 573$) gives +0.080 [+0.001, +0.162], Frame N ($n = 448$, O2r_{m30}) gives +0.117 [+0.020, +0.218], and the exploratory pool gives +0.096 [+0.034, +0.158]. (b) PSP for the six OPEN_{home} components on Frame N. NOV_{res} carries the signal (+0.208), while edge persistence is null (−0.013).

Frame N did NOT confirm the Cheng reversal ($-0.064 [-0.159, 0.031]$). Versus legacy newborns, Frame N concepts were 14% narrower, 89% more transient and 26% less sustained.

Pooling Frame N with Exp10 (exploratory): $+0.096 [+0.034, +0.158]$ at R3.

6.3 Experiment 14: Cheng’s consistency — size effect, not reach

Artifact workspace: `gen_art_experiment_14`.

Rebuilt Cheng et al.’s ideational consistency on 12,499 concepts. Test A (Cheng’s design, 105,839 concept-years): the negative-binomial twin reproduced $b = 0.428 (+53.5\%/SD)$. Adding $\log V(t)$ left $+1.3\% [+0.5, +2.1]$; A2/A1 ratio = $0.021 [0.009, 0.035]$. The volume effect was almost entirely a proxy for current size.

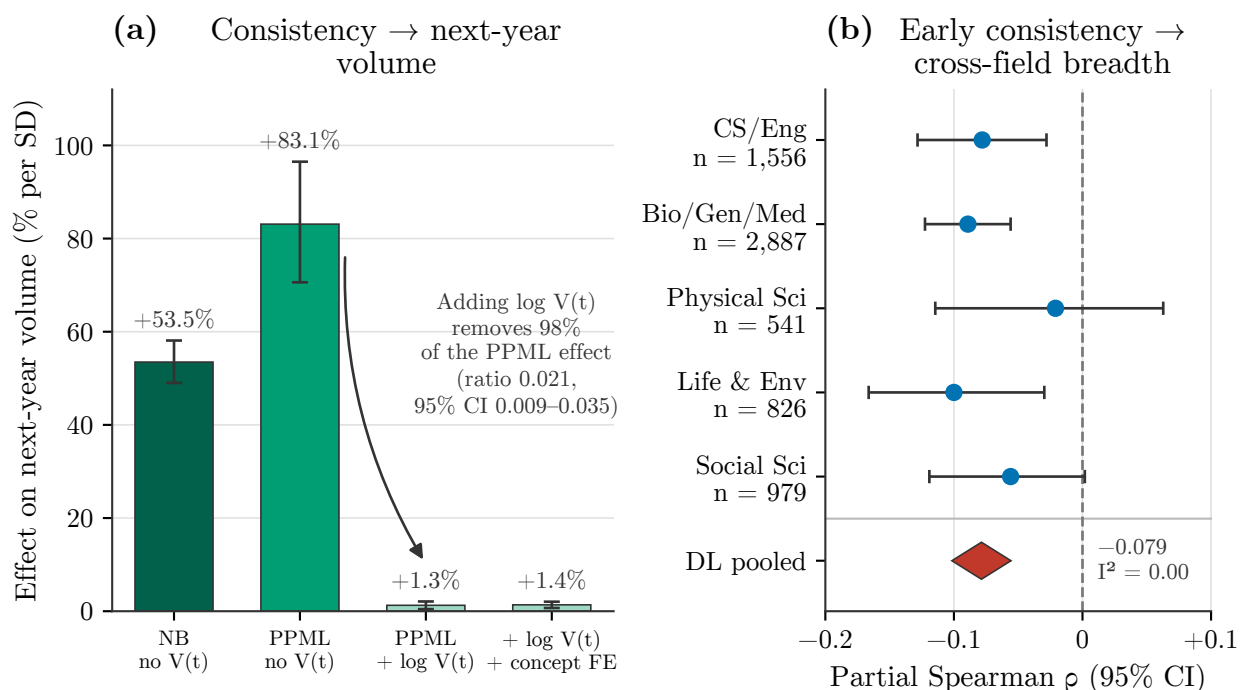


Figure 9: The consistency–breadth reversal. (a) Effect of ideational consistency on next-year volume: $+53.5\%$ without size control, $+1.3\%$ with $\log V(t)$. (b) Partial Spearman ρ between early consistency and rarefied cross-field breadth. All five group estimates are negative; DL pooled $-0.079 [-0.102, -0.056]$, $I^2 = 0$.

As an early trait, consistency predicted *narrower* cross-field reach. Net of B5, the partial Spearman with rarefied breadth was $-0.069 [-0.093, -0.047]$ (DL over 5 groups: -0.079 , $I^2 = 0$, 5/5 negative). Replicated on the 2015–2017 cohort: $-0.111 [-0.197, -0.030]$. Consistency had Spearman $+0.77$ with edge persistence — it was essentially weighted persistence.

Within concepts, a more consistent year was followed by slightly *more* entries ($b = +0.025$, $[0.004, 0.048]$), so the reach penalty was a between-concept trait. No Palla size $\times$ consistency interaction.

6.4 Experiment 15: Why churning concepts spread; Exp11 test completed

Artifact workspace: `gen_art_experiment_15`.

Three parts. **Part A** (partner decomposition): The home-only NOVCHURN signal (+0.118 pooled) came from new-community partners ($C_2 = +0.102$, Holm $p = 0.0025$) arriving through mixed-field papers ($C_4 = +0.103$, Holm $p = 0.0025$). Domain partners, not methodological partners, were the main contributor (domain Shapley $\phi = +0.132$ on the 2015–17 cohort, method = +0.012).

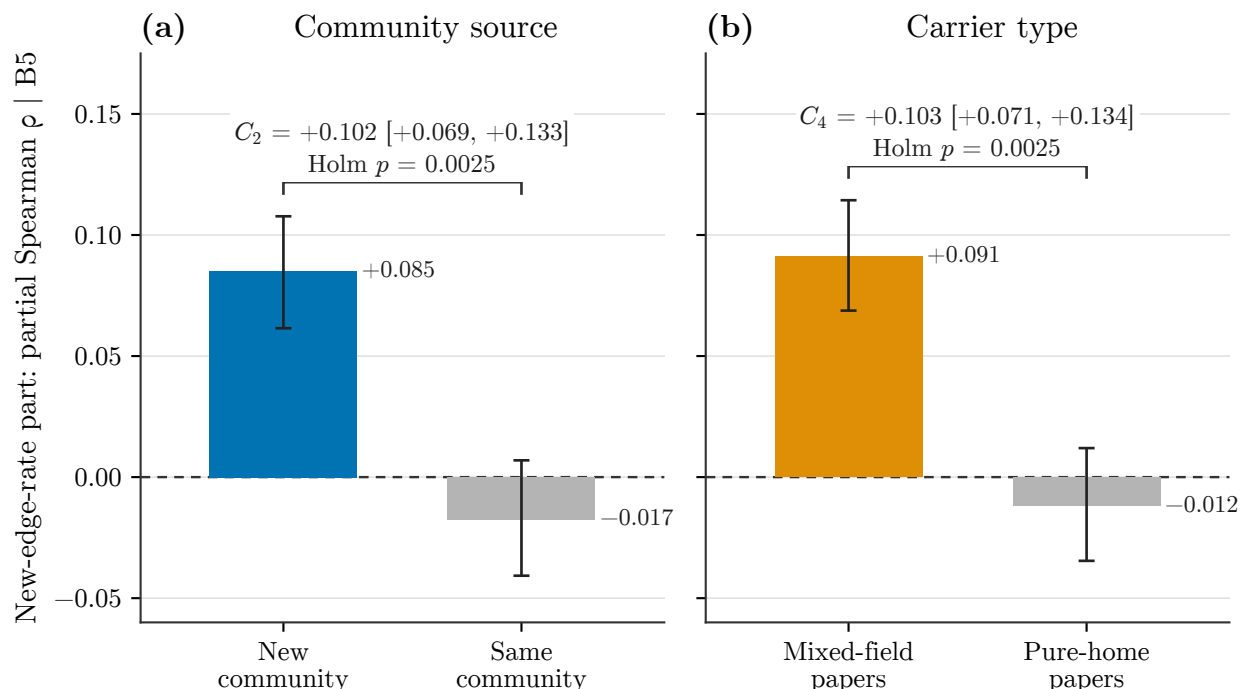


Figure 10: Which cooccurrence partners carry the openness–breadth signal? (a) Partners from new Leiden communities carry the signal ($C_2 = +0.102$ [+0.069, +0.133]); same-community partners do not. (b) Partners arriving through mixed-field papers carry it ($C_4 = +0.103$ [+0.071, +0.134]); partners from pure-home-field papers do not.

Bridging papers (5% of early home papers) halved the NOVCHURN partial from 0.118 to 0.056. These papers had more first-time authors (+5 percentage points) and far more off-home topics (+25 points).

Part B (trait stability): Prediction P-B1 FAILED. Yearly $OPEN_{home}$ ICC was 0.34–0.39; NOVCHURN 0.26–0.29. Openness was a fair trait measured through a noisy yearly window (dis-attenuated retest 0.86–0.91).

Part C (Exp11 completion): DEV verdict unchanged: NOT SUPPORTED. Old held-out PPML: $OPEN_{home} -0.079$ [−0.146, −0.013], opposite of the predicted sign. Cohort: both null. Sun–Abraham event study, DEV never-treated: -0.018 [−0.042, +0.004], pre-trend $p = 0.52$. H-S1 held pooled (+0.076 [+0.024, +0.126]) but not on OLD_HELDOUT. H-P1 as preregistered failed: the community half was +0.216 [+0.081, +0.351], the method half -0.055 .

6.5 Experiment 16: Is neighbourhood churn real or small-sample noise?

Artifact workspace: `gen_art_experiment_16`.

Confound check on 13,444 selection concepts. The within-concept year-label permutation null absorbed the temporal churn: excess NOVCHURN pooled PSP = +0.008 [−0.02, +0.03], compared

to raw $+0.116$ $[+0.09, +0.14]$. What predicted breadth was a static property — the non-redundancy of the concept’s home-topic partner mix — not year-to-year turnover. The R^2 of raw persistence on its permutation-null mean was 0.66.

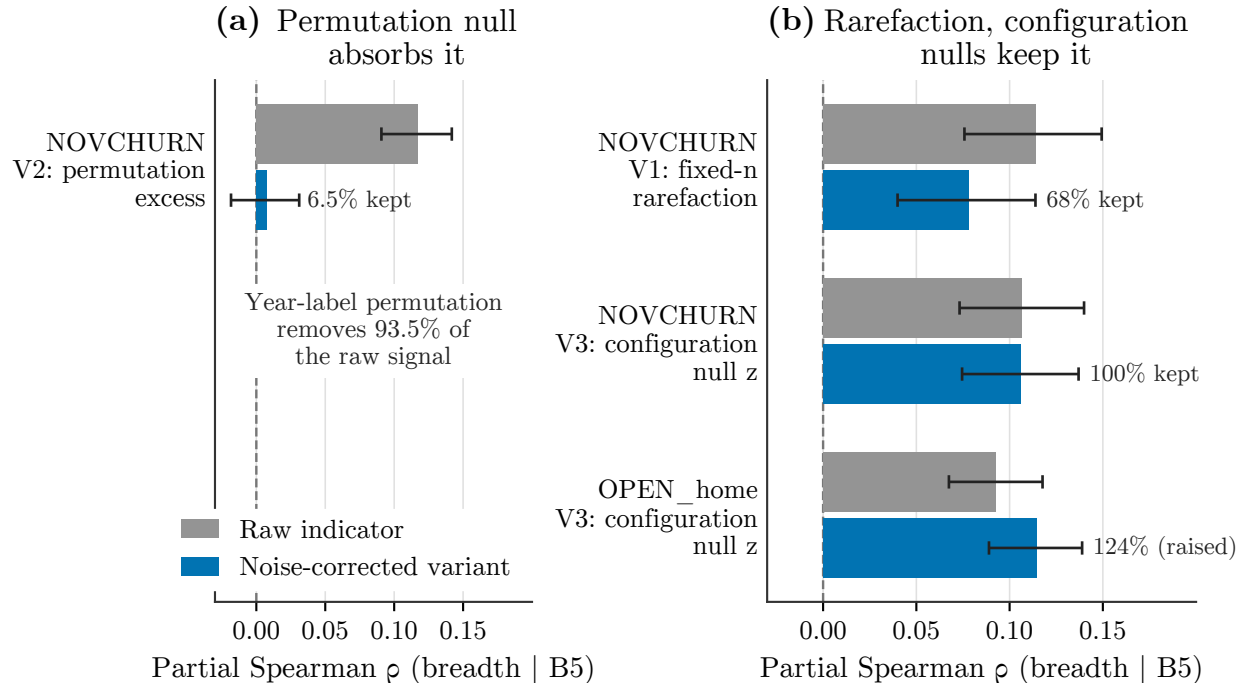


Figure 11: The churn/openness signal behaves like topical non-redundancy of the home neighbourhood, not year-to-year partner turnover. (a) The within-concept year-label permutation excess (V2) cuts NOVCHURN from $+0.117$ to $+0.008$, keeping 6.5%. (b) Fixed- n rarefaction (V1) keeps 68%; the configuration-null z -score (V3) keeps 100% and raises $\text{OPEN}_{\text{home}}$ to 124%.

Fixed-size rarefaction (10 papers/year) retained 68% of the pooled association. The configuration-null z -score (degree-preserving curveball) retained 100% and raised $\text{OPEN}_{\text{home}}$ from $+0.092$ to $+0.115$.

Frozen verdict: PARTLY THIN. V2 excess variants had split-half SB ~ 0.01 – 0.05 and the planted-churn check failed, so V2 could not adjudicate temporal churn at ~ 10 papers/year. Turnover was *untested*, not refuted. Split-half reliability: $\text{NOVCHURN}_{\text{raw}}$ 0.48, $\text{OPEN}_{\text{home}}$ 0.49. Disattenuated pooled $\text{NOVCHURN}_{\text{raw}}$ PSP: 0.178 $[0.141, 0.214]$ (approximate).

6.6 Evaluation 4: Record repair and openness evidence pool

Artifact workspace: `gen.art.evaluation.4`.

The final evaluation: zero new data, \$0 LLM. Claims ledger v4: 1,769 rows, 0 MISMATCH, 0 NOT_FOUND, 0 orphans.

Evidence synthesis (descriptive; R^2 , O2r_m50; DL on Fisher z + HKSJ): $\text{OPEN}_{\text{home}}$ non-selection pool (4 held-out groups + 2010–14 + 2015–17 cohorts, $k = 6$): $+0.069$ DL $[+0.038, +0.100]$, HKSJ $[+0.042, +0.096]$, $I^2 = 0$, 6/6 positive. DEV selection body $+0.109$, shrinkage 1.58. $\text{NOVCHURN}_{\text{home}}$ ($k = 5$): $+0.105$ $[+0.069, +0.140]$.

6.7 Review and Hypothesis Update

Review score: 2 (blocking). The submitted report was the paper manuscript rather than the chronological notebook. Conclusions contradicted artifacts (“monotone” dose-response used DEV values; “demonstrates” non-redundancy contradicted by V2’s inability to adjudicate; OPEN_{all} +0.17 headlined despite being mechanically coupled). Many dead ends were absent.

Final hypothesis update: the headline number was the Eval4 descriptive pool of OPEN_{home} over 6 non-selection bodies: +0.069 [0.038, 0.100], $I^2 = 0$, 6/6 positive. Evidence state: **lead**; confidence: **unchanged**.

7 Pre-Onset Footprint

About half of the end-of-window field density and volume-weighted field density signal came from the concept’s pre-onset footprint. On held-out data, the post-onset-only field density dropped from +0.374 to +0.187 [+0.145, +0.246], and the volume-weighted variant from +0.317 to +0.176 [+0.114, +0.227] (source: Evaluation 3, iteration 4). Post-onset D_vol was nearly rank-identical to B5 reach (ρ 0.97–1.00).

8 Retained-Field Relatedness and Field Entry

On the Exp7 independent frame (11,841 concepts, 6,978 entry events in 6,076 informative strata), the pooled held-out standardised coefficient was $d_0 = 0.322$ [0.291, 0.355] with LR = 325.8 ($p < 10^{-70}$). The effect was heterogeneous: Life & Environment showed +0.401, Mathematics & Decision Sciences +0.065 (null). DL pooled $d_0 = 0.243$ [0.118, 0.368], $I^2 = 0.92$.

Retained-field relatedness survived both the conventional RCA density (D_{rca}) and a share-weighted current-presence density. But the volume-matched contrast was null on held-out data (Holm $p = 0.76$), and under minimum-conditional-probability proximity, $d_0 = -0.021$. The frozen verdict was FRONTIER = PARTIAL.

A dose-response by retention age showed held-out values of 0.098 (2 years), 0.075 (3 years), 0.304 (≥ 4 years) — not monotone, though the signal was largest for long-retained fields.

Gateway centrality does not predict field retention. On the full panel of 27,393 episodes, gateway centrality added $\Delta\text{AUC} -0.00001$ [−0.0006, +0.0003]. The iteration-1 lead (+0.103) could not be certified above a shuffled- R placebo’s 95th percentile (0.130).

9 The Consistency–Breadth Reversal

Cheng et al.’s ideational consistency, rebuilt on OpenAlex topic co-usage for 12,311 concepts (105,839 concept-years), predicted next-year volume at +53.5%/SD in their design (no current-size control). Adding current volume $\log V(t)$ removed almost all of it: +1.3% [+0.5%, +2.1%], a residual-to-raw ratio of 0.021.

As an early trait, consistency predicted *narrower* cross-field reach: partial Spearman −0.069 [−0.093, −0.047] on the pooled selection bodies (DL over 5 groups: −0.079, $I^2 = 0$, 5/5 negative), replicated on the 2015–2017 cohort (−0.111 [−0.197, −0.030]). Not confirmed on Frame N (−0.064 [−0.159, 0.031]).

10 Breadth Decomposition

The exact decomposition of the top-vs-bottom O_{2r} tercile gap showed exploration contributing 73% and retention 27% (variant i_pooled, DEV). The pre-registered variant iv (Medicine excluded) gave DL 0.504 [0.329, 0.679], $I^2 = 0.76$. The frontier advance component was negative ($s_M = -0.046$): broad concepts start with wider contact, not by advancing a wider frontier. These shares are an accounting identity, not causal effects.

11 Mechanism: Which Partners Carry the Signal

The home-only signal was carried by partners from new Leiden communities arriving through mixed-field papers. New-community partners: $C_2 = +0.102$ [+0.069, +0.133], Holm $p = 0.0025$. Mixed-field papers: $C_4 = +0.103$ [+0.071, +0.134], Holm $p = 0.0025$. Novelty among substantive-domain partners, not methodological partners, was the main contributor (domain Shapley $\phi = +0.132$, method = +0.012).

Controlling for the share of early home papers that were bridging papers halved the NOVCHURN partial from 0.118 to 0.056.

12 Topical Non-Redundancy, Not Temporal Turnover

The within-concept permutation null absorbed the year-to-year churn component. The V2 excess (temporal component above the permutation null) was indistinguishable from zero: pooled PSP +0.008 [-0.02, +0.03], vs raw +0.116 [+0.09, +0.14]. Fixed-size rarefaction retained 68%, and the degree-preserving configuration null retained 100%.

However, V2 excess variants had split-half reliability of only ~ 0.01 – 0.05 , so V2 could not adjudicate temporal churn at the available sample sizes (~ 10 papers/year). Turnover was *untested*, not refuted.

13 Dead Ends and Negative Results

1. **Naturalisation gap** A_h^* (Exp1, iteration 1): failed every pre-registered clause. $\Delta\rho = -0.006$.
2. **D_ratio and F_res** (Exp3, iteration 1): no candidate survived.
3. **G landing on O_{2r}** (Exp4, iteration 1): $\Delta\rho = +0.033$, failed.
4. **Gateway centrality predicts field retention** (Exp5, iteration 2): held-out $\Delta\text{AUC} = -0.00001$. DISCONFIRMED at scale.
5. **H3 gateway-weighted landing on breadth** (Exp5, iteration 2): pooled CI [-0.006, 0.065] includes zero.
6. **Rescue and relay** (Exp6, iteration 2): NOT SUPPORTED on held-out.
7. **Ordering** (Exp6, iteration 2): MIXED. 57/87 non-tied = 65.5%; but negative lead-lag coefficients, significant pre-trend, and the reverse path significant on dev.
8. **Two trajectory classes** (Exp6, iteration 2): overturned by Exp12, iteration 4 (HMM-vs-DTW ARI = 0.094).

9. **G-variant O1 gains** (Eval1, iteration 2): label-coverage artefacts ($G +0.072 \rightarrow +0.002$).
10. **Abandonment penalty** (Exp7, iteration 3): inconclusive. $d_{\text{lost}} A1 = -0.007 [-0.036, 0.022]$; in R4 with d_0 , it turned positive (+0.064).
11. **O5 external recognition** (Exp8, iteration 3; Dataset 2, iteration 2): no indicator or model beat B5 + onset year. Only 42% of positives marked a genuinely new concept.
12. **Experiment 9** (iteration 3): never ran (output-format validation failed). The RQ2 trajectories artifact.
13. **Experiment 2** (iteration 1): never ran (REPL stalled). Candidate S co-authorship.
14. **Candidate S co-author indicators** (Exp8, iteration 3): S_comp, S_comp_n, S_isolated_share computed and scored; not confirmed for any outcome.
15. **Experiment 11 closure test** (iteration 4, completed iteration 5): NOT SUPPORTED on DEV; opposite-signed on old held-out (-0.079).
16. **PR2 “localised keep more early”** (Exp12, iteration 4): REVERSED on DEV (-0.110).
17. **n_authors_early non-replication** (Exp10, iteration 4): does not replicate for O3 (+0.014) or O1b (+0.036) on the fresh cohort.
18. **RETENTION_RATIO_early** (Exp10, iteration 4): does not survive concept-type controls on the fresh cohort (R2 $-0.043 [-0.116, 0.031]$).
19. **P-B1 trait prediction** (Exp15, iteration 5): FAILS. Yearly OPEN_{home} ICC 0.34–0.39.
20. **Frame N Cheng reversal** (Exp13, iteration 5): NOT confirmed ($-0.064 [-0.159, 0.031]$).
21. **Persistence-filtered RCA density rival** (D_rca_persist_k): differs from Exp7’s D_rca_pers (max $\rho = 0.877$); the exact rival from Research 2 remains untested.

14 Evidence Synthesis

The Eval4 descriptive pool of OPEN_{home} over 6 non-selection bodies (4 held-out groups + 2010–14 cohort + 2015–17 cohort; $k = 6$) gave:

Table 9: Evidence synthesis: OPEN_{home} partial Spearman with O_{2r} ($m = 50$) at R2 across non-selection bodies. Source: Evaluation 4, iteration 5.

Estimator	PSP	95% CI	I^2
DerSimonian–Laird	+0.069	[+0.038, +0.100]	0.00
HKSJ	+0.069	[+0.042, +0.096]	0.00
DEV (selection body)	+0.109	—	—
NOVCHURN _{home} ($k = 5$)	+0.105	[+0.069, +0.140]	—

All 6 bodies were positive. DEV shrinkage was 1.58. The shuffled-feature placebo pool was +0.021 $[-0.010, +0.051]$, which included zero.

Table 10: Run bookkeeping. Five iterations, 20 artifacts commissioned, 16 completed, 4 failed or incomplete. Review scores and moves from the iteration records. No trajectory log with ledger spend was available for this run.

Iter.	Move	Review	Executed	Key artifacts
1	Wide screen (5 rivals)	3 (blocking)	3/5	Exp1, Exp3, Exp4
2	Deepen gateway lead	3 (blocking)	5/5	Exp5, Exp6, Eval1, Dataset2, Res1
3	Frontier test + RQ1 screen	3 (blocking)	4/5	Exp7, Exp8, Eval2, Res2
4	Fresh cohort + RQ2	2 (blocking)	4/5	Exp10, Exp12, Eval3, Res3
5	Frame N + Cheng + confound	2 (blocking)	5/5	Exp13, Exp14, Exp15, Eval4, Exp16

15 Run Bookkeeping

16 What the Run Learned Overall

16.1 Confirmed findings

1. **Seven early indicators predict cross-field breadth on held-out fields.** From six families, 53 indicators were screened against a five-feature popularity baseline. Seven survived held-out validation (Table 5). Four belong to the retained-frontier family and three to cooccurrence topology. No centrality or volume-only indicator survived.
2. **The OPEN index captures topical non-redundancy.** $\text{OPEN}_{\text{home}}$ predicts breadth on the 2015–2017 confirmatory cohort (PSP +0.091 at R2, marginal at R4/R5) and on vocabulary-free Frame N (PSP +0.117 at R3, PARTIAL verdict). The Eval4 non-selection pool is +0.069 [+0.038, +0.100], $I^2 = 0$, 6/6 positive.
3. **The consistency–breadth reversal.** Cheng et al.’s ideational consistency predicts volume (+53.5%/SD raw, +1.3% with size control) but predicts *narrower* cross-field reach (−0.069, DL −0.079, $I^2 = 0$, 5/5 negative). Replicated on the 2015–17 cohort (−0.111); not confirmed on Frame N.
4. **Concepts spread next to fields related to those retaining them.** On the independent frame, $d_0 = 0.322$ [0.291, 0.355]. Verdict: FRONTIER = PARTIAL (volume-matched contrast null; reverses under min-cp proximity).
5. **Breadth is 73% early exploration.** The decomposition shows early contact diversity, not frontier advance or retention, drives the breadth gap.
6. **Background homophily explains two-thirds of raw lineage variance.** $R^2 = 0.66$ of the raw lineage log-odds ratio on the background log-odds ratio (Exp1, iteration 1).

16.2 Effect sizes and limitations

The OPEN–breadth association is modest. $\text{OPEN}_{\text{home}}$ adds no forecasting gain over the baseline (+0.002 [−0.003, +0.008]). Split-half reliability is 0.49. The legacy concept vocabulary is survivor-selected (Frame N concepts are 14% narrower and 89% more transient). The signal is not temporal turnover (the permutation null absorbs it), but turnover could not be tested at available sample sizes (V2 excess reliability ~ 0.01 –0.05). Life & Environment Sciences show the weakest signal.

17 What Is Still Open

1. **Temporal turnover vs static dispersion.** Exp16's V2 excess is unreliable ($SB \sim 0.01-0.05$); the planted-churn check fails. Whether temporal partner turnover adds anything beyond static neighbourhood diversity remains untested.
2. **The persistence-filtered RCA density rival.** Research 2's `D_rca_persist_k` differs from Exp7's `D_rca_pers` ($\max \rho = 0.877$); the exact rival remains untested.
3. **Proximity dependence of the field-entry effect.** Under Hidalgo's minimum-conditional-probability proximity, d_0 reverses (-0.021). The retained-frontier claim holds only on the sparse PMI backbone.
4. **Domain heterogeneity.** Life & Environment Sciences show the weakest OPEN signal (PSP $+0.071$ vs $+0.186$ for the other five units pooled). The retained-field relatedness effect has $I^2 = 0.92$. No moderator was found.
5. **The Cheng reversal on Frame N.** Not confirmed (-0.064 , CI includes zero).
6. **Causal mechanism.** Early ego-network openness could reflect the concept's intrinsic generality, the diversity of the research community around it, or the breadth of the problems it addresses. The run showed association, not causation.